Milestone 3 Help Session

Building the Qt AI Workspace, one piece at a time

Lecture 14 | Fall 2026 | M3 due Oct. 24

Plan for the entire three weeks

Today

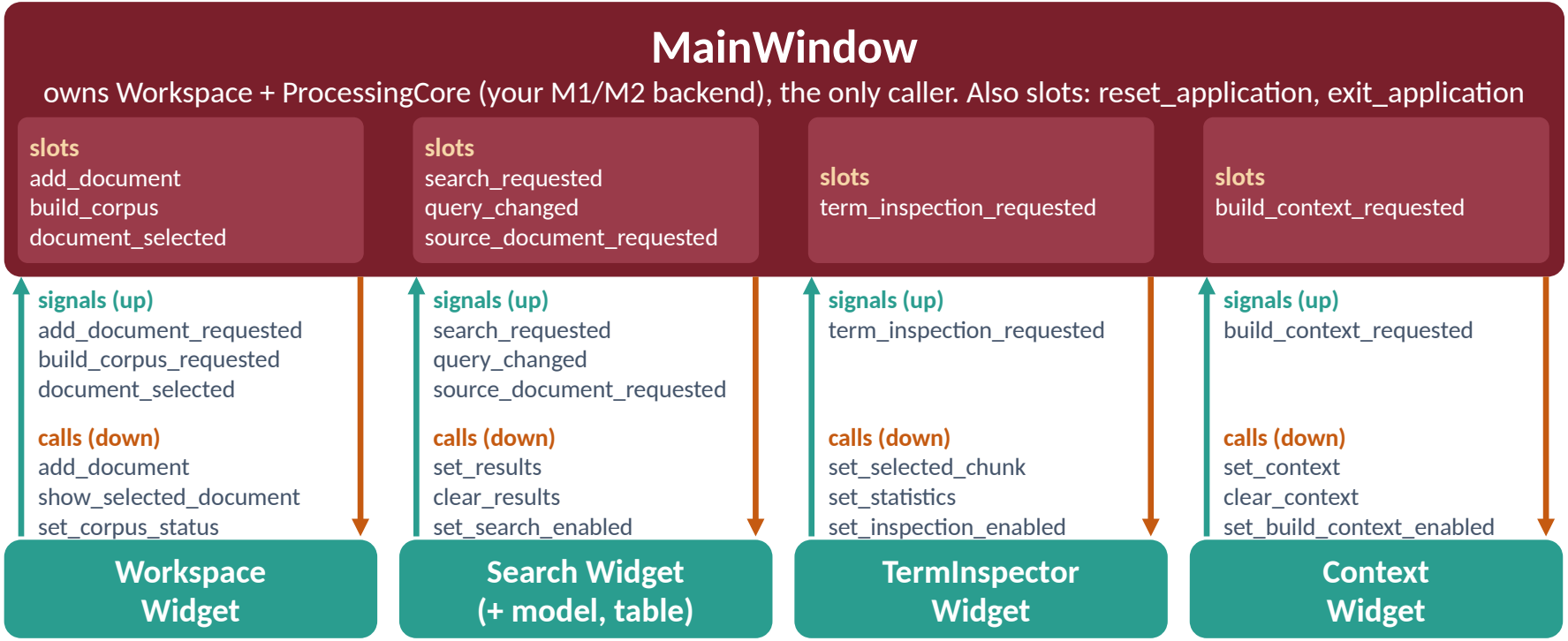
- Architecture: who does what
- One event, start to finish
- Keeping state consistent
- Debugging tips
- Group activity: Spot the State Bug

Suggested checkpoints

- Week 1: window, layouts, startup state, model + table with fake rows
- Week 2: Add, Build, Search, View Source
- Week 3: Inspector, Context, Reset, Exit, state fixes

Run the program after every step. Do not wire everything at once.

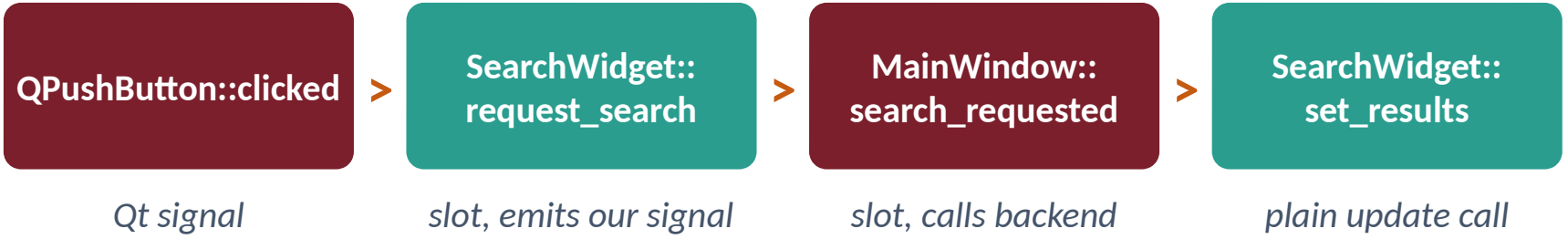
Architecture: who does what



The backend finds it, the model holds it, the view shows it.

Note: Search's selected_chunk_changed connects straight to TermInspector's set_selected_chunk, skipping MainWindow. (the only shortcut)

One click, start to finish: Search



Use: &Class::member

```
// connect(sender, &SenderClass::signal,
//         receiver, &ReceiverClass::slot);
connect(search_button_,
        &QPushButton::clicked,
        this, &SearchWidget::request_search);
```

Not allowed: a lambda

```
connect(search_button_,
        &QPushButton::clicked,
        this, [this]() {
            request_search();
        });
```

No lambdas: the slot is named with &Class::member. Every M3 event follows this same path.

Startup state and layout rules

Startup (and Reset) state

- Corpus: Not built
- Results k = 3
- Token budget = 300
- Build Corpus, Search, Build Context, Inspect, View Source: disabled
- Status: Ready

Layout rules

- Open at 900 x 900, stay resizable
- No setFixedSize(), no x/y positions: use layouts
- Long text in QPlainTextEdit so it scrolls
- Non-white color: header, Reset, Exit

Model / View: SearchResultModel

Model = data. View = display. The backend does the search.

Rank	Document	Chunk	Score	Text
1	doc_ai.txt	4	0.812345	Signals and slots …

```
// What the view asks the model:
//   rowCount()     how many rows? (0 for a valid parent)
//   columnCount()  how many columns?
//   data()         what text goes in this cell?

// When the stored results change:
beginResetModel();
    // replace the stored results
endResetModel();       // the view redraws
```

Classic mistakes: columnCount() still 0, no reset calls, search logic inside the model.

Wiring the table in SearchWidget

```
result_model_ = new SearchResultModel(this);
results_view_ = new QTableView(this);
results_view_->setModel(result_model_);          // 1. model first
results_view_->setSelectionBehavior(QAbstractItemView::SelectRows);
results_view_->setEditTriggers(QAbstractItemView::NoEditTriggers);

connect(results_view_->selectionModel(),         // 2. then this
        &QItemSelectionModel::selectionChanged,
        this, &SearchWidget::show_selected_result);
```

- selectionModel() exists only after setModel(); before that it is null
- In the slot: result_at(row), then check for nullptr
- The table is read-only and selects whole rows

Events 1 and 2: Add Document, Build Corpus

Add Document

- Pick a file (Cancel: change nothing)
- Read it; report problems in the status line
- Update the backend and the display
- The data changed: what is now out of date?

Build Corpus

- One backend call: rebuild
- Mark it: a corpus now exists
- Clear it: old results, details, stats, context
- Update it: status text, enabled buttons

Think about the order of updates: does changing one place trigger events that read another? Building is always explicit: adding a document never rebuilds.

Events 3 and 4: Search, then select a result

Search

- MainWindow: ProcessingCore::search(query, k)
- Hand the results to SearchWidget; clear the old context
- SearchWidget: replace the table, clear selection and details, disable View Source
- Zero results is valid

Select a result

- Get the row; result_at(row); check nullptr
- Show details, enable View Source
- Tell the app which chunk is selected
- View Source: emit the document id; MainWindow selects it in the list

Selecting never runs a new search. Think: what else depended on the old selection?

Term Inspector and Context

Term Inspector

- Empty term: no request
- DF needs only a current corpus
- Chunk TF only if a result is selected

Context

- Slider 50 to 1000, step 50, starts at 300
- Uses current query, k, and budget
- Not the selected row

Backend calls (from M2)

MainWindow uses ProcessingCore::document_frequency, term_frequency and build_context. Moving a slider or editing a field never runs them.

State consistency: what is enabled when

One function decides: update_action_states()

Control	Enabled when
Build Corpus	the Workspace has at least one document
Search	corpus is current AND the query has non-whitespace text
Build Context	same as Search
Inspect	corpus is current
View Source	a valid result row is selected (inside SearchWidget)

Call it after every event that changes one of these: add, build, query edit, reset.

State consistency: what each event changes

Event	Results	Details	Term stats	Context	Corpus current
Add Document	cleared	cleared	cleared	cleared	no
Build Corpus	cleared	cleared	cleared	cleared	yes
Edit query / k / budget	kept	kept	kept	kept	same
Search	replaced	cleared	cleared	cleared	same
Select result	kept	shown	see spec	kept	same
Reset	cleared	cleared	cleared	cleared	no

Editing an input is not requesting an operation: it never runs processing. Details: spec, section on state consistency.

Reset and Exit

Reset: back to startup

- Clear the backend Workspace (not only the list)
- Mark the corpus as not current
- Ask each of the four components to reset itself
- Recompute the enabled buttons
- One call, four behaviors: virtual dispatch

```
// main.cpp: Exit callback
void confirm_exit(QWidget* parent) {
    // QMessageBox::question(parent, ...)
    // Yes -> QApplication::quit();
    // No  -> just return
}
window.set_exit_callback(&confirm_exit);
```

Remember

- ExitCallback = void (*)(QWidget*)
- MainWindow's Exit slot calls it (check for null)
- No: state stays unchanged

Debugging Qt: symptom, then first check

Symptom	Check first
Button does nothing	Did you call connect()? Did you emit the signal?
undefined reference to vtable	moc did not run: Q_OBJECT, AUTOMOC ON, re-run cmake
Crash at startup	Null widget pointer: create widgets before using them
Table stays empty	columnCount() 0? Reset calls? data() for DisplayRole?
Wrong button state, stale output	update_action_states() after that event? invalidate?
Window grows with long text	Use QPlainTextEdit (scrolls), not QLabel

Class activity: Spot the State Bug

Group + individual

Four short bugs, each breaking a state rule from today. Find them, fix them, explain what the user would see.

1. Think (1 min)

Read the bugs silently.

2. Discuss (6 min)

Groups of 3-4. Explain your reasoning out loud.

3. Answer (4 min)

Write YOUR OWN answers in Canvas.

Groups may share ideas; every student submits an individual answer.

Bugs 1 and 2

Discuss in groups

Bug 1: the new document shows a blank display

```
workspace_widget_->add_document(title);
workspace_.add_document(Document(id, title, text));
invalidate_corpus();
```

Bug 2: after Reset, Build Corpus is still enabled

```
void MainWindow::reset_application() {
    for (auto* part : parts) part->reset();
    update_action_states();
}
```

Hint for Bug 1: adding a list row also selects it, which fires document_selected(row).

Bugs 3 and 4

Discuss in groups

Bug 3: typing in the query box runs a search

```
void MainWindow::query_changed(const QString& q) {
    search_requested(q, search_widget_->result_count());
}
```

Bug 4: the Term Inspector keeps an old chunk after results are cleared

```
void SearchWidget::clear_results() {
    result_model_->clear();
    selected_result_text_->clear();
    view_source_button_->setEnabled(false);
}
```

Think: after a search, what does SearchWidget tell the rest of the app? Does clear_results() do the same?

Final answers: on your own

Individual

1. Fix it

Choose one bug. Write the corrected code, or the missing lines.

2. What the user sees

One sentence: what goes wrong for the user without your fix?

3. Why

In 1-2 sentences: why does MainWindow call update_action_states() after state changes?

Before you submit

Checklist

- Opens at 900 x 900, still resizable
- Startup/reset: Not built, k = 3, budget = 300
- Buttons enabled only when valid
- Add Document invalidates the corpus
- Walk through the full example session

Help

- TA office hours
- Reference package on Canvas (read only)
- M3 due Oct. 24